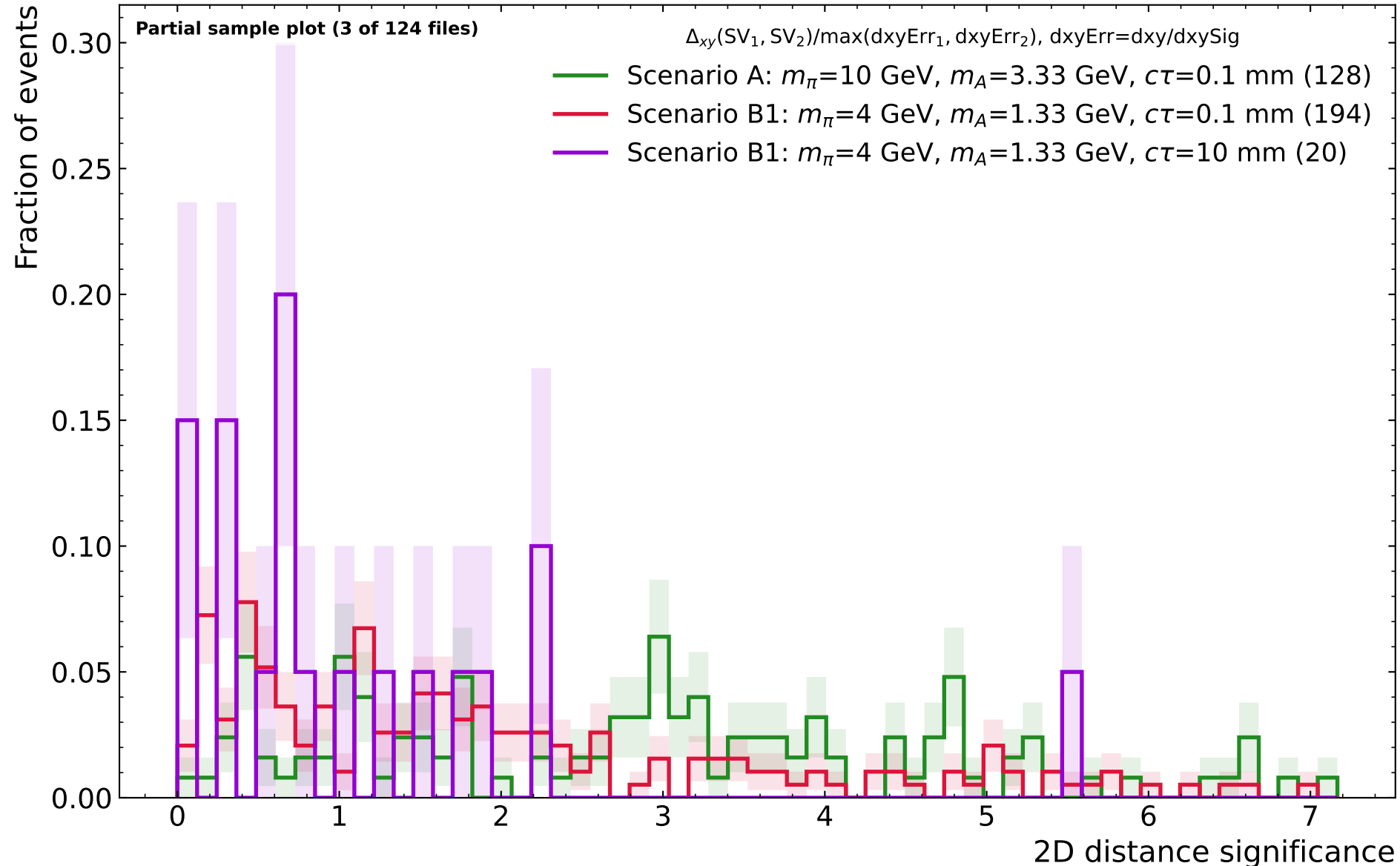


quadv: fourmuonSV + 2 consistent muonSV
trigger (Mu10 OR DoubleMu) + offline kinematic pre-selection
each distribution is normalised to unit area; bands/bars are statistical only

CMS Simulation Preliminary

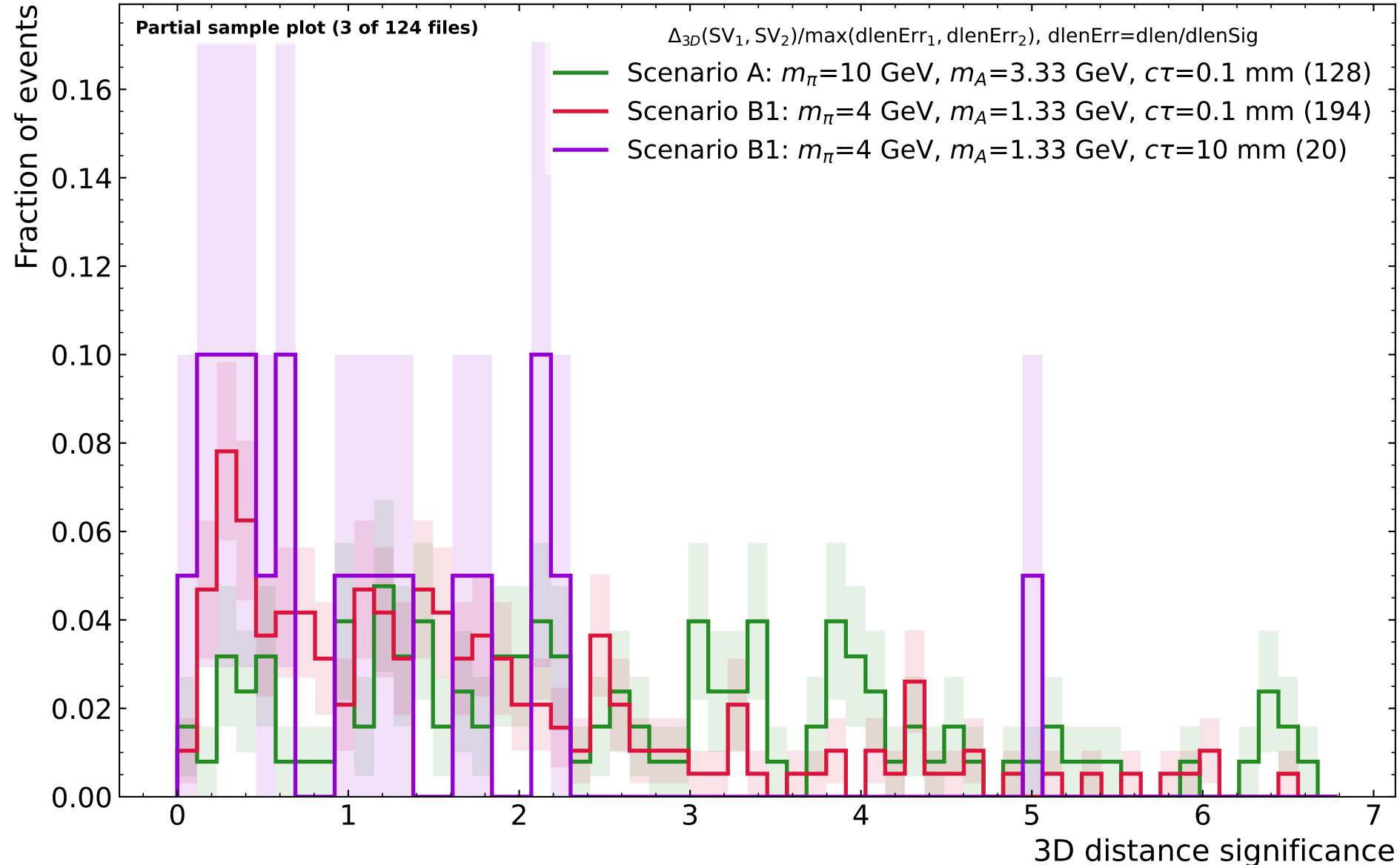
(13.6 TeV, 2024)



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CMS Simulation Preliminary

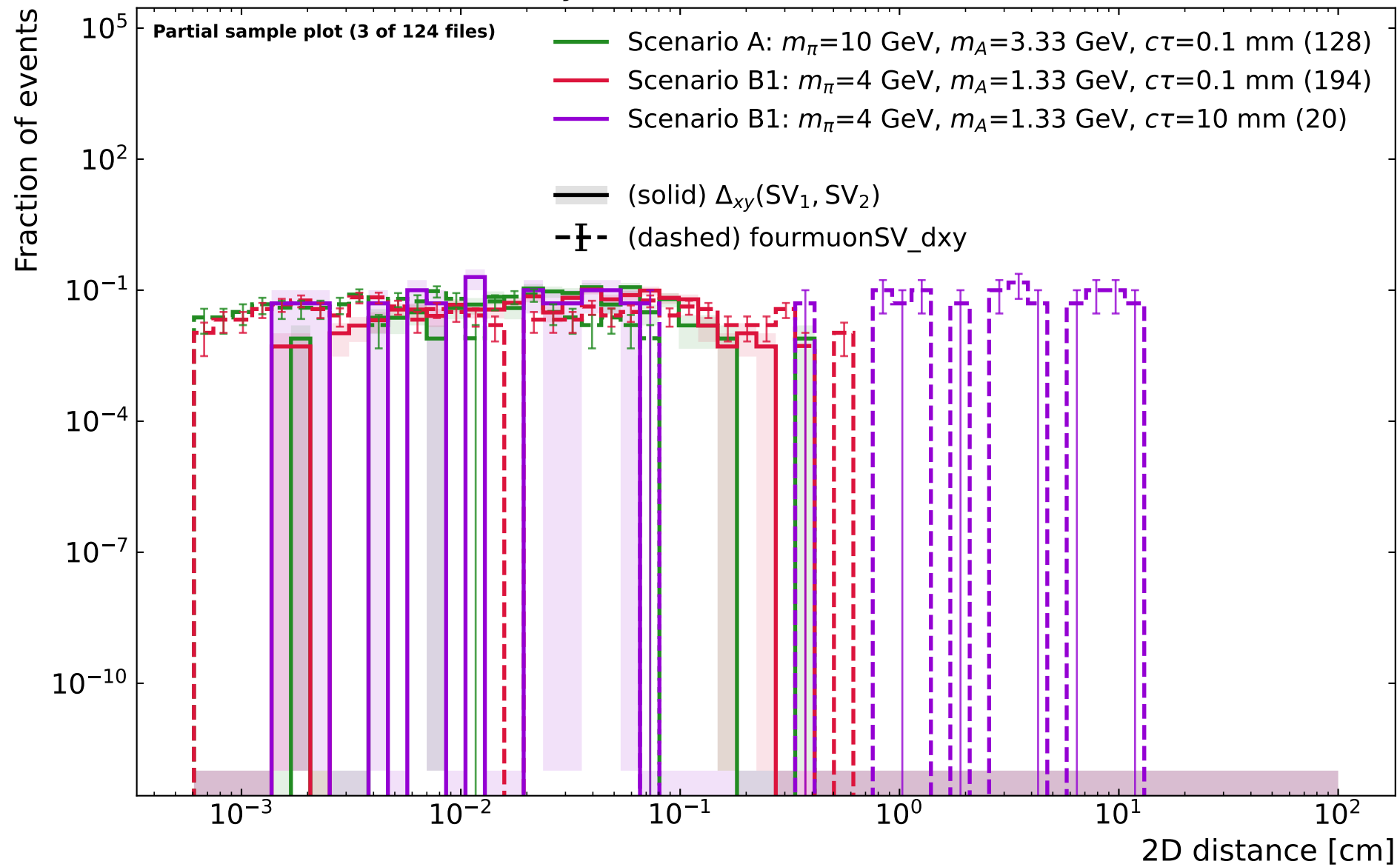
(13.6 TeV, 2024)



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 trigger (Mu10 OR DoubleMu) + offline kinematic pre-selection
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CMS *Simulation Preliminary*

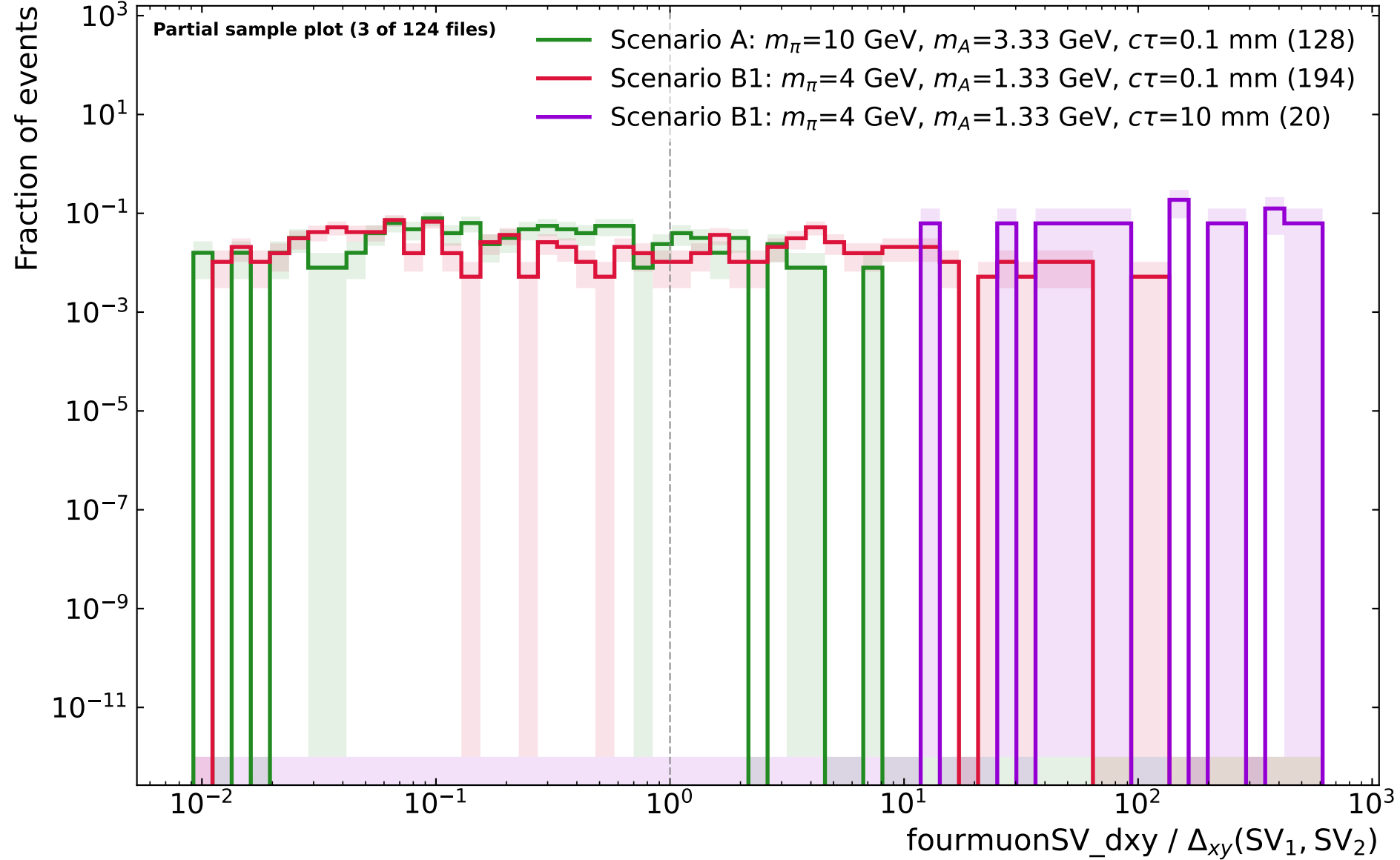
(13.6 TeV, 2024)



quadv: fourmuonSV + 2 consistent muonSV
trigger (Mu10 OR DoubleMu) + offline kinematic pre-selection
each distribution is normalised to unit area; bands/bars are statistical only

CMS *Simulation Preliminary*

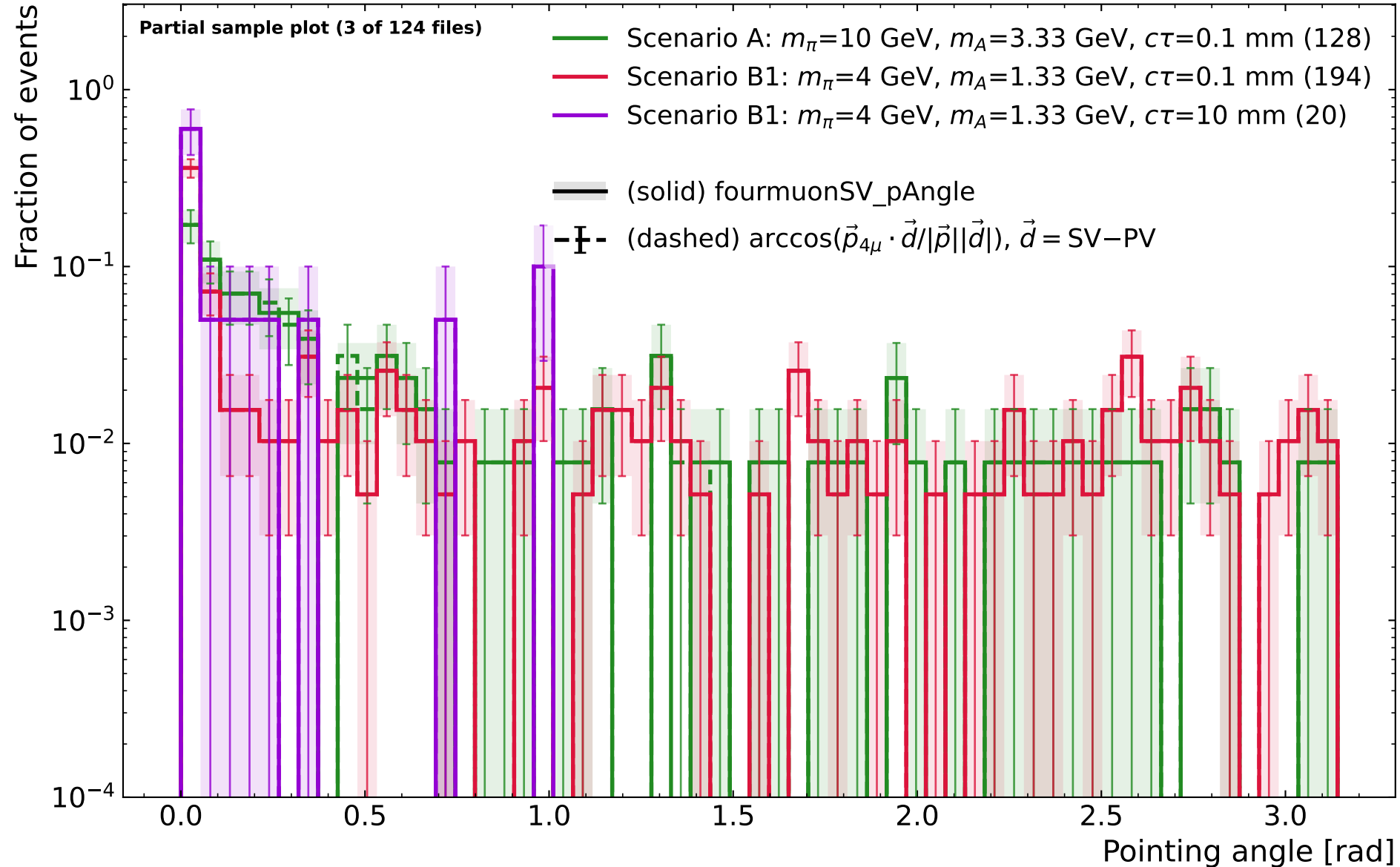
(13.6 TeV, 2024)



quadv: fourmuonSV + 2 consistent muonSV
 trigger (Mu10 OR DoubleMu) + offline kinematic pre-selection
 each distribution is normalised to unit area; bands/bars are statistical only

CMS Simulation Preliminary

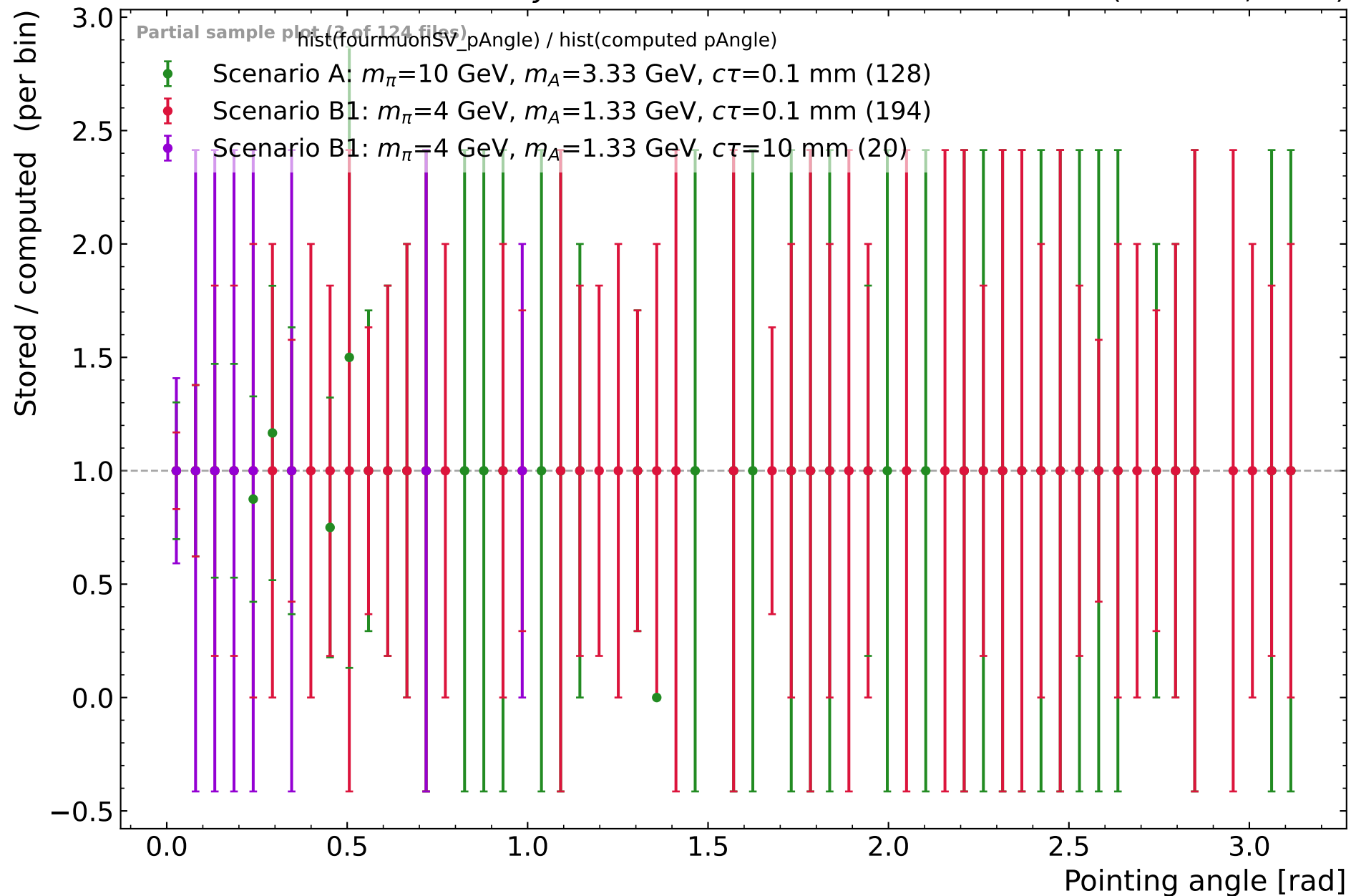
(13.6 TeV, 2024)



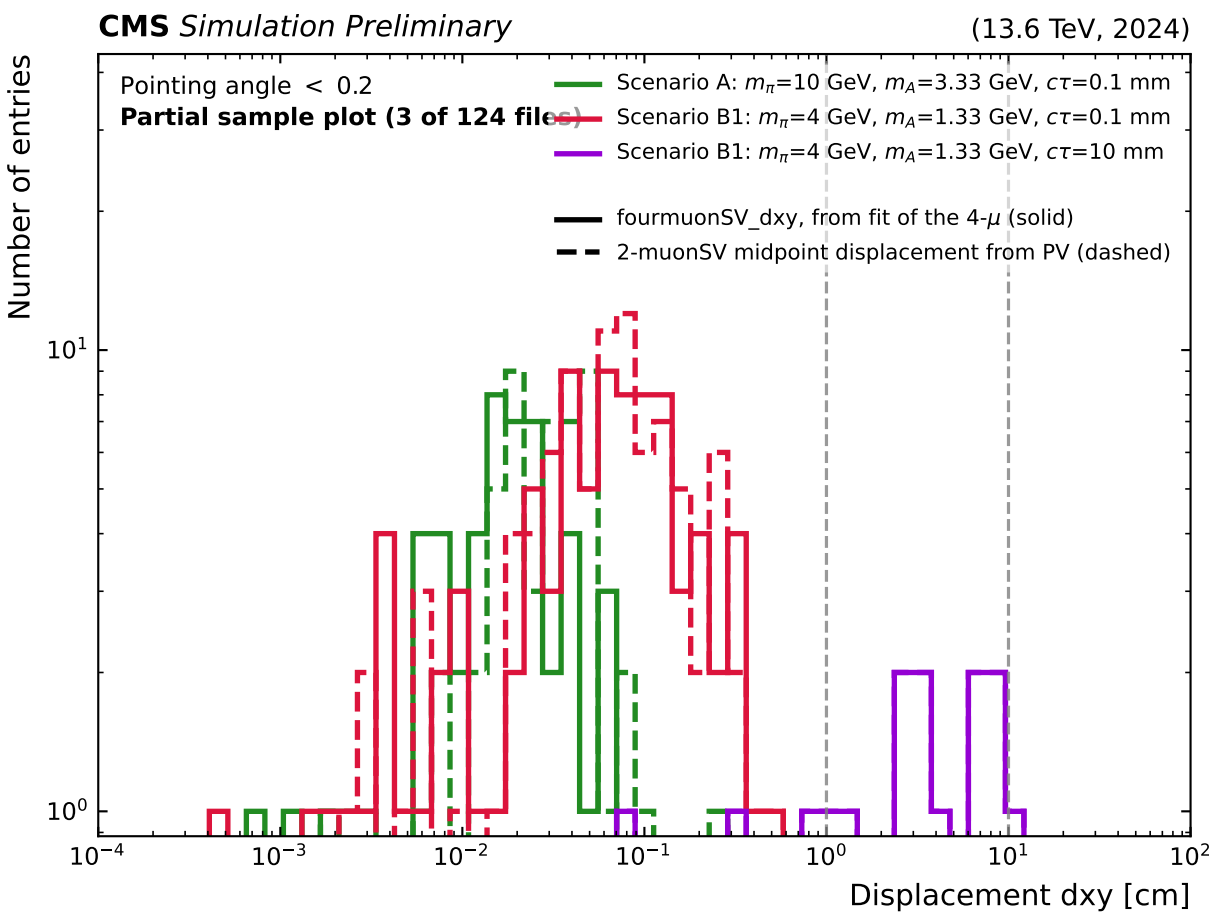
quadv: fourmuonSV + 2 consistent muonSV
trigger (Mu10 OR DoubleMu) + offline kinematic pre-selection

CMS *Simulation Preliminary*

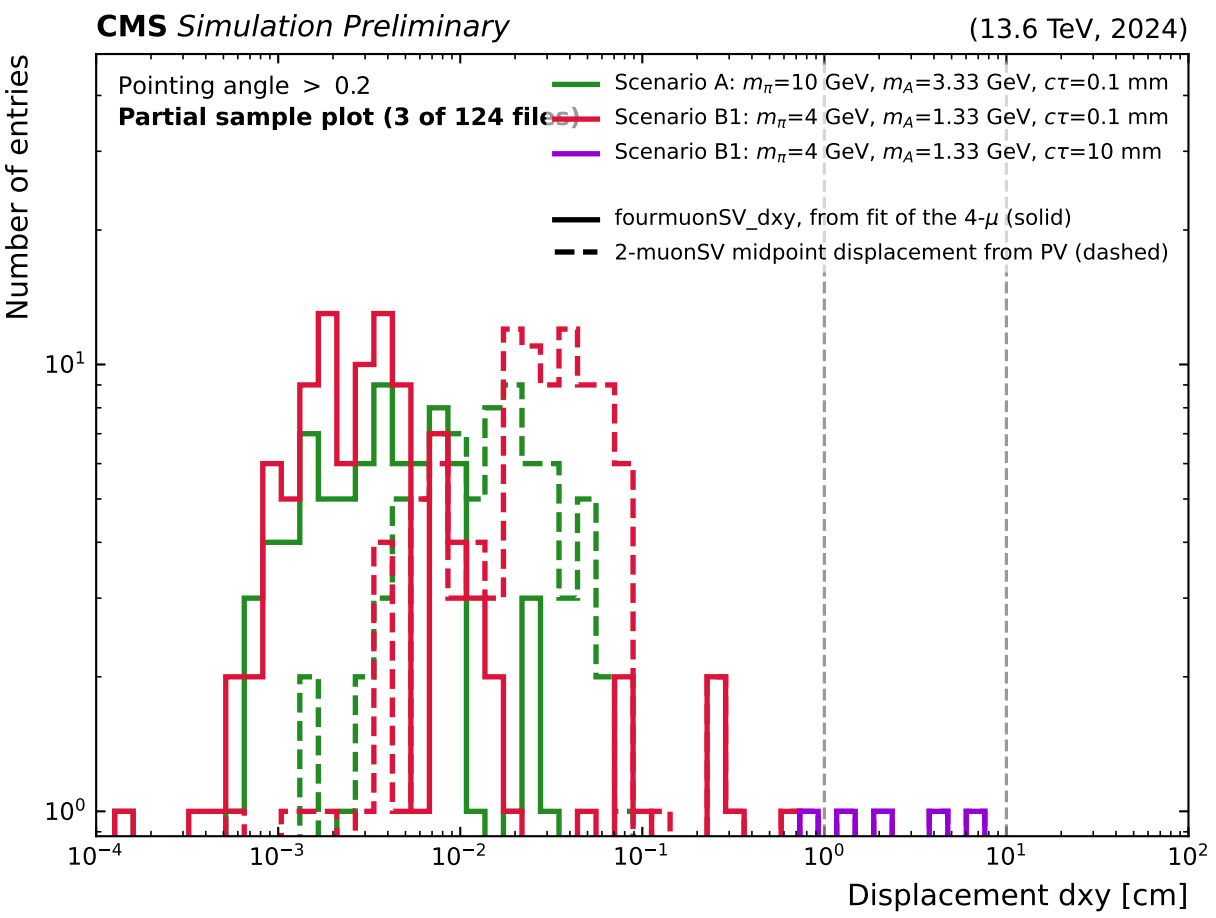
(13.6 TeV, 2024)



quadv category populations vs dxy, split by fourmuonSV_pAngle
solid: fourmuonSV_dxy, from fit of the 4- μ dashed: 2-muonSV midpoint displacement from PV grey lines: dxy category edges (1, 10 cm)



	dxy < 1	1--10	dxy > 10	Total
A $c\tau=0.1$ fourmuonSV	51	0	0	51
A $c\tau=0.1$ 2muSV mid	51	0	0	51
B1 $c\tau=0.1$ fourmuonSV	90	0	0	90
B1 $c\tau=0.1$ 2muSV mid	90	0	0	90
B1 $c\tau=10$ fourmuonSV	3	11	1	15
B1 $c\tau=10$ 2muSV mid	3	11	1	15



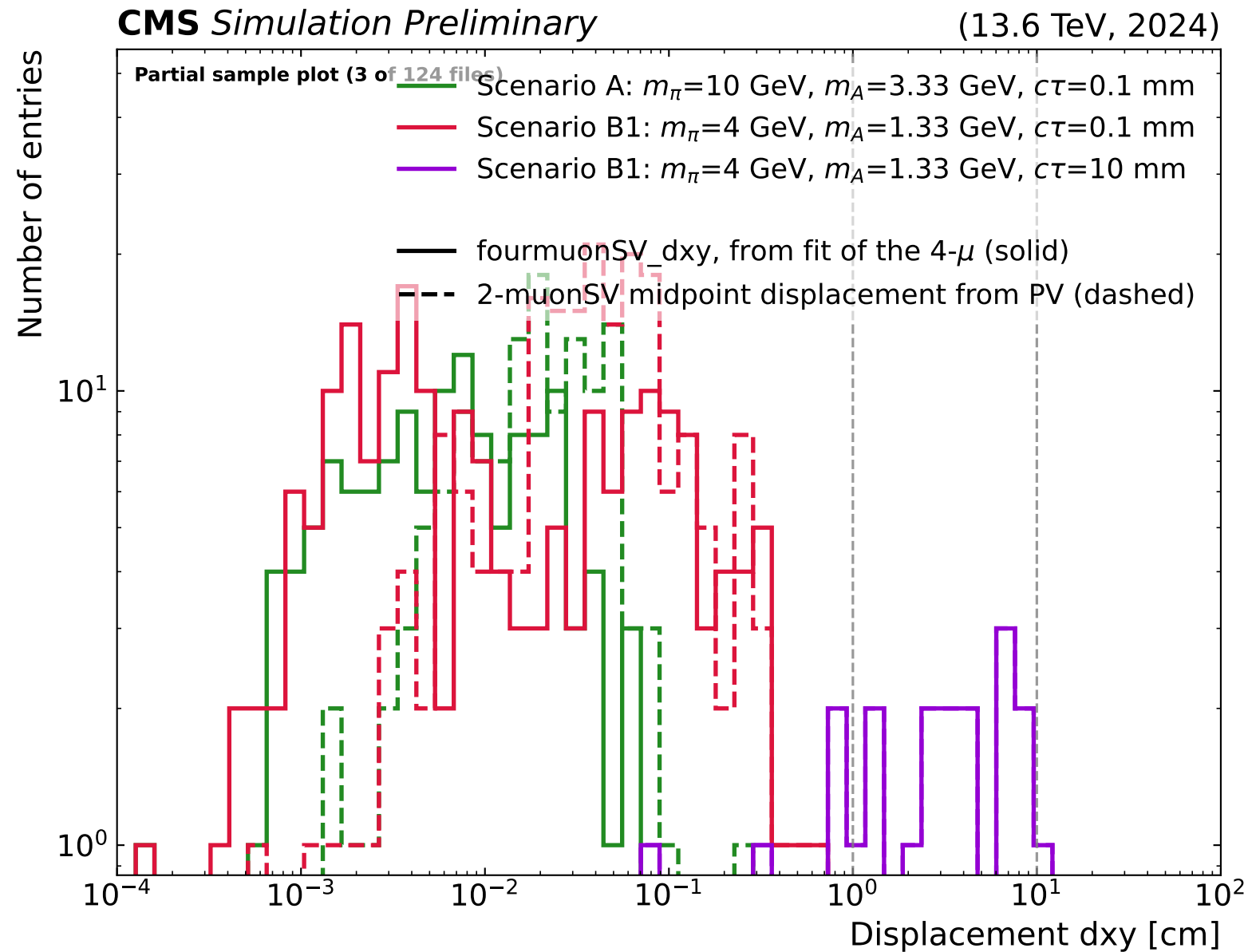
	dxy < 1	1--10	dxy > 10	Total
A $c\tau=0.1$ fourmuonSV	77	0	0	77
A $c\tau=0.1$ 2muSV mid	77	0	0	77
B1 $c\tau=0.1$ fourmuonSV	104	0	0	104
B1 $c\tau=0.1$ 2muSV mid	104	0	0	104
B1 $c\tau=10$ fourmuonSV	1	4	0	5
B1 $c\tau=10$ 2muSV mid	1	4	0	5

quadv category populations vs dxy (pAngle-inclusive)

solid: fourmuonSV_dxy, from fit of the 4- μ

dashed: 2-muonSV midpoint displacement from PV

grey lines: dxy category edges (1, 10 cm)



	dxy < 1	1--10	dxy > 10	Total
A $c\tau=0.1$ fourmuonSV	128	0	0	128
A $c\tau=0.1$ 2muSV mid	128	0	0	128
B1 $c\tau=0.1$ fourmuonSV	194	0	0	194
B1 $c\tau=0.1$ 2muSV mid	194	0	0	194
B1 $c\tau=10$ fourmuonSV	4	15	1	20
B1 $c\tau=10$ 2muSV mid	4	15	1	20

Ratio of dxy distributions: fourmuonSV_dxy / (2-muonSV midpoint dxy)
grey dashed: dxy category edges (1, 10 cm); dotted: ratio = 1

